Suraj Sreeprakash Menon

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EXPERIENCE

- **Forward Deployed Engineer Intern** – *Plivo* Feb 2026 – Sep 2026
  - Shipped an **n8n** + **Python** automation that pulls a customer's call and message history from Redshift on ticket creation and posts an LLM summary to the thread, removing the manual log lookup that preceded every triage (**∼20 hrs/month** of support time).
  - Debugged production voice and messaging failures across Plivo Voice, SMS, WhatsApp, Verify, Zentrunk SIP trunking, and PHLO, driving root cause analysis with application logs, **OpenSearch**, **Wireshark** PCAPs, and Postman replay.
  - Instrumented customer traffic monitoring in **Grafana** (calls-per-second, concurrency trends) to catch capacity and rate-limit issues before they surfaced as outages.
  - Primary technical point of contact for enterprise customers across **APAC, EMEA, and the US**, resolving **200+ tickets/month** and partnering with Engineering, DevOps, and Product to close platform bugs, feature requests, and outages.
- **Full Stack Developer Intern** – *Jabsz Gaming Studios* Oct 2025 – Feb 2026
  - Led a full UI/UX redesign of the company website, fixing cross-page inconsistencies and establishing one consistent visual theme across all pages.
  - Migrated the production frontend from **React** to **Next.js (App Router)**, integrating legacy JavaScript and global CSS assets via custom Next.js configuration and build tooling.
  - Migrated the production database from **Firebase** to **Supabase**, porting data into a relational PostgreSQL schema and rewriting all client-side data access against the new SDK.
  - Improved performance and SEO by converting all images to WebP and adopting `next/image`, raising Lighthouse scores from **∼50 to ∼90**.
- **Full Stack Developer Intern** – *BharatCrest Pvt. Ltd.* Jun 2025 – Aug 2025
  - Built **HRMatcher**, a recruitment platform that screens candidates with skill-based filtering and similarity ranking via the Google Gemini API, processing 100+ resumes in under 5 minutes.
  - Developed a full-stack restaurant management system (**React** + **FastAPI**) with real-time order tracking, billing, and automated inventory management with low-stock alerts.
  - Added route-level code splitting and image lazy-loading, cutting client-side load times **20–30%** in Lighthouse audits.

PROJECTS

- **Tether** – *Swift, SwiftUI, Network.framework, CryptoKit*
  - Mac host + iPhone client for browsing shared folders, sending files, running a scoped terminal, and driving allowlisted coding agents (Claude Code, Cursor, Codex) as a live PTY – locally, with no account and no cloud.
  - Pairs via QR with a Keychain-held device key; every session is ChaChaPoly-encrypted over **Network.framework** with zero third-party networking or crypto libraries. Shipped as v3.3 with public release notes.
- **AI Test Case Generator** – *FastAPI, Next.js, Kafka, PostgreSQL, MinIO, Prometheus/Grafana*
  - Built an LLM system that turns SRS documents and UI screenshots into structured test cases, with a Next.js frontend and a FastAPI REST backend.
  - Kafka background workers decouple upload from generation; MinIO stores source documents, PostgreSQL tracks job state, and Prometheus + Grafana monitor queue depth and worker health.
- **NanoPrune Canvas** – *PyTorch, ONNX, FastAPI, Next.js, React Flow*
  - Browser toolkit that simulates magnitude pruning (FP32 → INT4) and post-training quantization before edge deployment, parsing PyTorch/ONNX models into an interactive graph with live per-layer sparsity.
  - Hardware panel reports flash/SRAM fit, inference time, and energy per inference across **26 embedded targets**, so model fit is known before deployment rather than after.
- **Saizen** – *Swift, Next.js, Capacitor, AVPlayer, Keychain*
  - Personal iOS media client whose playback sources are installable modules resolved at runtime – a broken source ships as a **module update, not an app release**.
  - Custom AVPlayer chrome with gesture seeking and skip marks, a resilient pause/resume offline download queue, and OAuth tokens stored in Keychain behind a fail-closed secret scan of every IPA.

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Bash, Swift
Frontend: React, Next.js, Vite, Tailwind CSS, SwiftUI, Recharts
Backend & Infra: FastAPI, Node.js, REST APIs, RabbitMQ, Kafka, SIP/Voice APIs
Data & Cloud: PostgreSQL, Amazon Redshift, OpenSearch, Supabase, Firebase, SQLite, MinIO, Vercel
AI & ML: Gemini API, LLM automation pipelines, Isolation Forest, PyTorch, ONNX
Tools: Git, GitHub Actions, Docker, Prometheus, Grafana, Wireshark, Postman

EDUCATION

- **Dayananda Sagar College of Engineering** Bengaluru, KA
  B.E. in Information Science & Engineering (VTU); CGPA: 8.9 *2022 – 2026*

PUBLICATION

- **Suraj Menon** et al., "NammaLakes: IoT-Based Real-Time Distributed Lakes Monitoring System," *International Journal of Latest Technology in Engineering Management and Applied Science*, 2025. [DOI] [Source]